

UK Railway Performance Analysis

1. Project Planning & Management

1.1 Project Proposal

This project focuses on analyzing railway operations and performance in the United Kingdom using a structured dataset of train activities.

The railway system in the UK, operated under organizations such as **National Rail**, is a critical component of transportation infrastructure. Efficient railway performance is essential for ensuring customer satisfaction, operational efficiency, and economic productivity.

The main problem addressed in this project is the lack of clear insights into train delays, route performance, ticket trends, and passenger flow when working with raw data.

The objective of this project is to develop an interactive dashboard using **Microsoft Power BI** to analyze railway data and provide insights into performance metrics, delays, and operational trends.

The dataset includes information such as train journeys, delays, ticket types, routes, and timing details.

1.2 Timeline & Team Roles

Phase	Task	Responsible	Duration
Phase 1	Planning & Understanding Dataset	All Members	Week 1
Phase 2	Data Cleaning & Preparation	Data Team	Week 1–2
Phase 3	Data Modeling	BI Developers	Week 2
Phase 4	Dashboard Development	BI Team	Week 3
Phase 5	Testing & Validation	All Members	Week 3
Phase 6	Documentation & Presentation	All Members	Week 4

1.3 KPIs

- Total Number of Trips
- Total Revenue (if ticket price exists)
- Average Delay Time

- On-Time vs Delayed Trips (%)
 - Most Frequent Routes
 - Peak Travel Times
 - Ticket Type Distribution
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2. Literature Review

2.1 Domain Background

Railway systems are complex networks that require continuous monitoring and optimization. In the UK, rail transport plays a vital role in daily commuting and logistics.

Data analytics helps railway operators understand delays, passenger demand, and operational inefficiencies. Tools like **Microsoft Power BI** enable visualization of large datasets, making it easier to detect patterns and improve performance.

2.2 Related Work

- Transportation analytics dashboards using **Microsoft Power BI**
 - Public transport delay analysis projects
 - Data visualization studies focusing on route optimization and time efficiency
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3. Requirements Gathering

3.1 Business Questions

- How many train trips are completed over time?
 - What is the average delay duration?
 - Which routes experience the most delays?
 - What are the busiest travel times (hours/days)?
 - Which ticket types are most used?
 - What percentage of trains are on time vs delayed?
 - Are there patterns in delays based on time or location?
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3.2 KPIs & Metrics

KPI	Definition
Total Trips	Count of all journeys
Average Delay	Average delay time per trip
Delay Rate	Percentage of delayed trips
On-Time Rate	Percentage of on-time trips
Top Routes	Most frequent routes
Peak Hours	Highest traffic times
Ticket Distribution	Usage of ticket types

3.3 Data Sources

Source	Description
railway.csv	Main dataset containing trip details
data_dictionary.csv	Description of dataset columns

Expected Columns (based on dataset type):

- Trip ID
- Departure Station
- Arrival Station
- Departure Time
- Arrival Time
- Delay Time
- Ticket Type
- Price (if available)

Estimated Size:

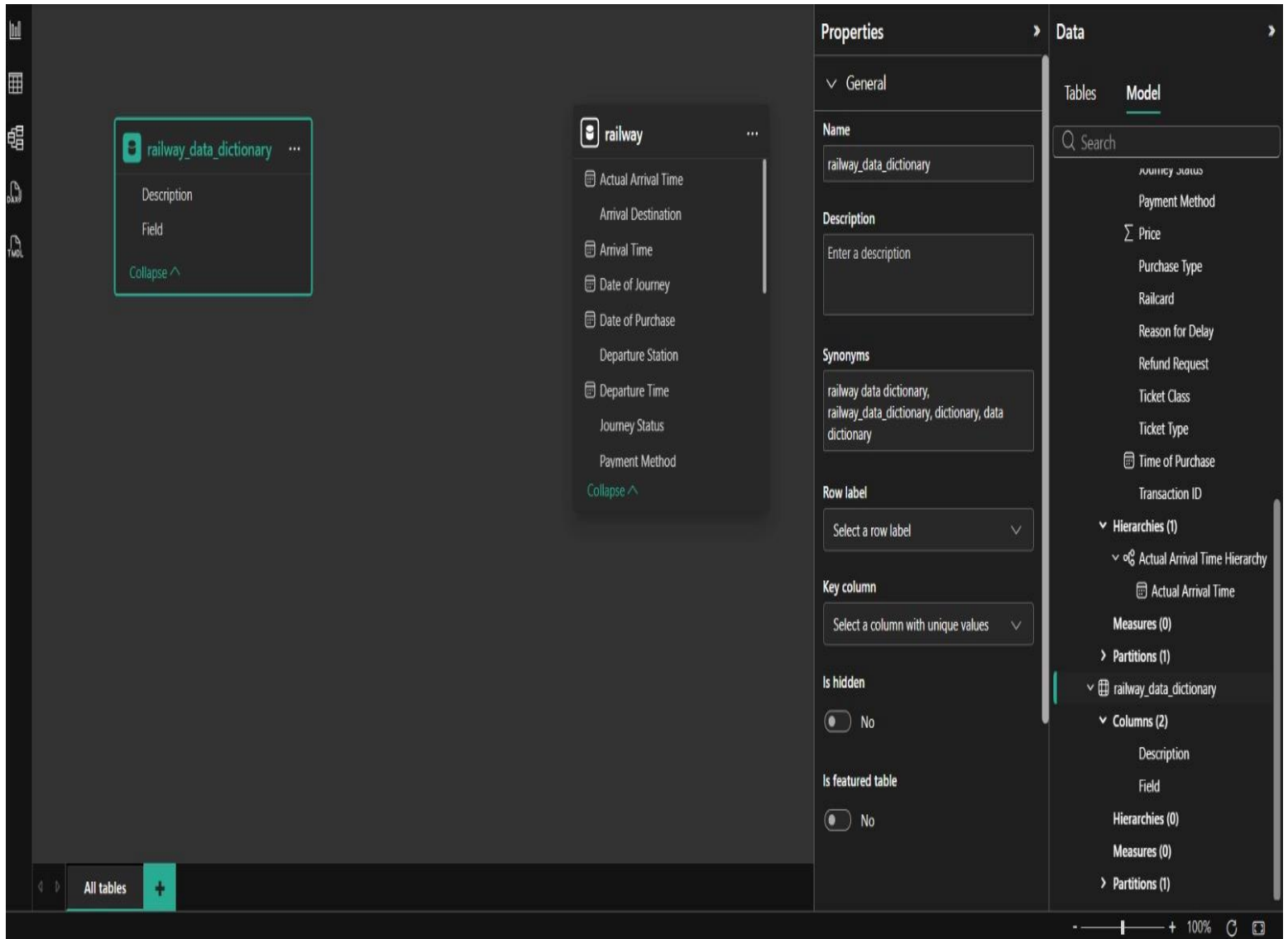
- Rows: 10,000+
- Key Columns: Trip ID, Route, Time, Delay, Ticket Type

4. System Analysis & Design

4.1 Data Model Diagram (Star Schema)

Description:

The data model is structured using a Star Schema, where the central fact table contains trip data, connected to dimension tables such as time, station, and ticket type.



The screenshot displays the Microsoft Power BI Desktop interface, specifically the 'Model' view. The central workspace shows a table named 'railway_data_dictionary' with a list of fields: Description, Field, and Collapse. The 'Properties' pane on the right provides details for the selected table, including its name, description, synonyms, row label, key column, and whether it is hidden or featured. The 'Data' pane on the right shows the table's structure, including columns and hierarchies.

Properties

- General**
 - Name: railway_data_dictionary
 - Description: Enter a description
 - Synonyms: railway data dictionary, railway_data_dictionary, dictionary, data dictionary
 - Row label: Select a row label
 - Key column: Select a column with unique values
 - Is hidden: No
 - Is featured table: No

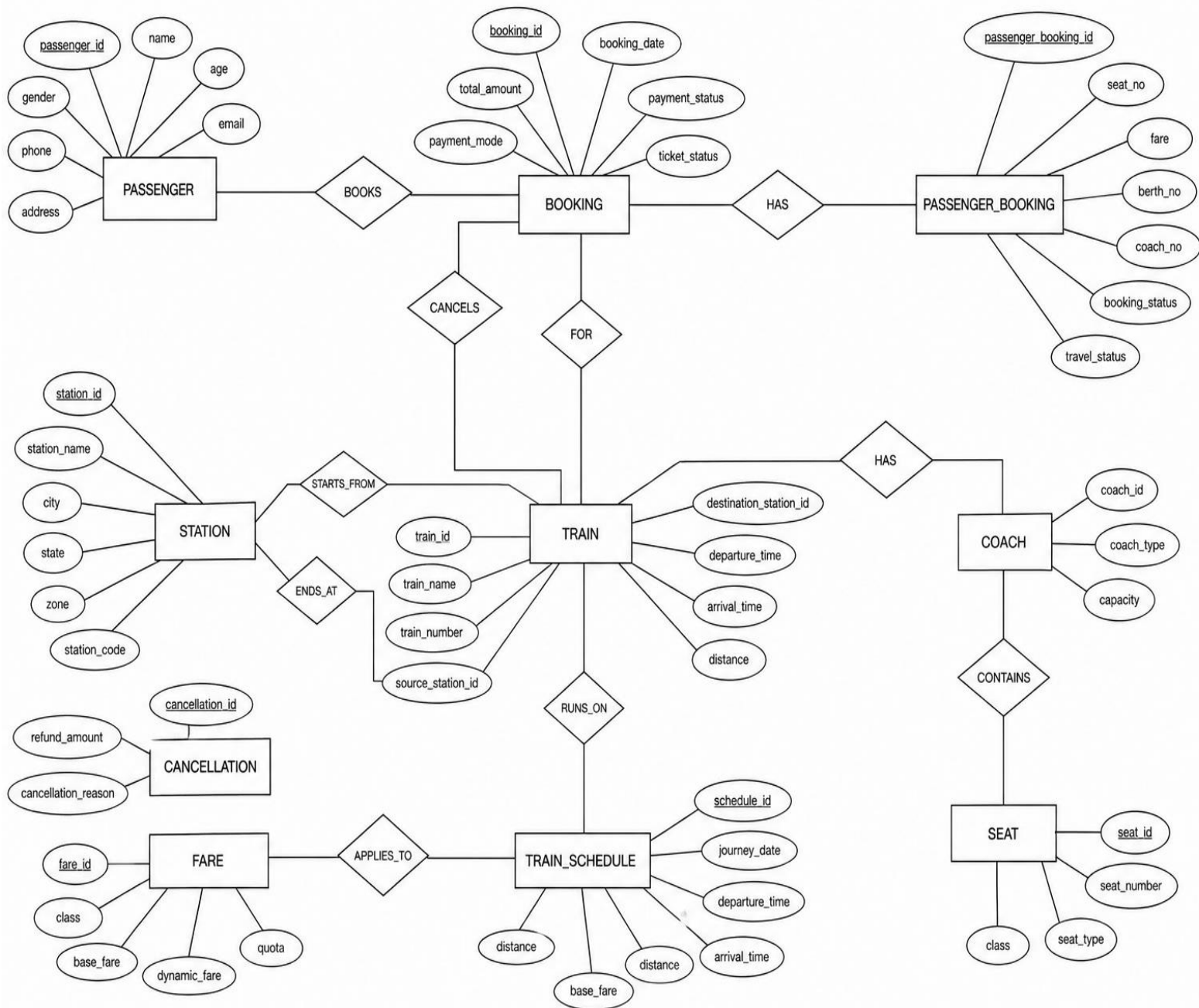
Data

- Tables**
 - Search
 - Actual Arrival Time
 - Payment Method
 - Price
 - Purchase Type
 - Railcard
 - Reason for Delay
 - Refund Request
 - Ticket Class
 - Ticket Type
 - Time of Purchase
 - Transaction ID
- Hierarchies (1)**
 - Actual Arrival Time Hierarchy
 - Actual Arrival Time
- Measures (0)**
- Partitions (1)**
 - railway_data_dictionary
- Columns (2)**
 - Description
 - Field
- Hierarchies (0)**
- Measures (0)**
- Partitions (1)**

4.2 Entity Relationship Diagram (ERD)

Description:

The ERD shows relationships between tables such as Trips, Stations, Time, and Tickets, including primary and foreign keys.



4.3 Data Cleaning Steps (Power Query)

The following steps were applied to prepare the dataset:

- Removed duplicate rows
- Handled missing or null values
- Converted time columns to proper datetime format
- Created calculated columns (e.g., delay duration)
- Standardized station names
- Filtered invalid records

= Table.TransformColumnTypes(#"Removed Errors",{"Departure Time", type datetimezone}, {"Arrival Time", type datetimezone}, {"Date of									
		Departure Time	Arrival Time	Actual Arrival Time	Journey Status	Reason for Delay	Refund Request		
		100% 0% 0%	100% 0% 0%	100% 0% 0%	100% 0% 0%	100% 0% 0%	100% 0% 0%		
		Valid Error Empty	Valid Error Empty	Valid Error Empty	Valid Error Empty	Valid Error Empty	Valid Error Empty		
1	1/1/2024	12/31/1899 11:00:00 AM +02:00	12/31/1899 1:30:00 PM +02:00	12/31/1899 1:30:00 PM +02:00	On Time	unknown	No		
2	1/1/2024	12/31/1899 9:45:00 AM +02:00	12/31/1899 11:35:00 AM +02:00	12/31/1899 11:40:00 AM +02:00	Delayed	Signal Failure	No		
3	1/2/2024	12/31/1899 6:15:00 PM +02:00	12/31/1899 6:45:00 PM +02:00	12/31/1899 6:45:00 PM +02:00	On Time	unknown	No		
4	1/1/2024	12/31/1899 9:30:00 PM +02:00	12/31/1899 10:30:00 PM +02:00	12/31/1899 10:30:00 PM +02:00	On Time	unknown	No		
5	1/1/2024	12/31/1899 4:45:00 PM +02:00	12/31/1899 7:00:00 PM +02:00	12/31/1899 7:00:00 PM +02:00	On Time	unknown	No		
6	1/1/2024	12/31/1899 6:15:00 AM +02:00	12/31/1899 8:05:00 AM +02:00	12/31/1899 8:05:00 AM +02:00	On Time	unknown	No		
7	1/1/2024	12/31/1899 10:30:00 PM +02:00	12/31/1899 11:40:00 PM +02:00	12/31/1899 11:40:00 PM +02:00	On Time	unknown	No		
8	1/1/2024	12/31/1899 12:00:00 AM +02:00	12/31/1899 12:30:00 AM +02:00	12/31/1899 12:30:00 AM +02:00	On Time	unknown	No		
9	1/1/2024	12/31/1899 12:00:00 AM +02:00	12/31/1899 1:50:00 AM +02:00	12/31/1899 2:07:00 AM +02:00	Delayed	Signal Failure	No		
10	1/1/2024	12/31/1899 1:30:00 AM +02:00	12/31/1899 2:30:00 AM +02:00	12/31/1899 2:30:00 AM +02:00	On Time	unknown	No		
11	1/1/2024	12/31/1899 1:45:00 AM +02:00	12/31/1899 2:35:00 AM +02:00	12/31/1899 2:35:00 AM +02:00	On Time	unknown	No		
12	1/1/2024	12/31/1899 2:15:00 AM +02:00	12/31/1899 3:15:00 AM +02:00	12/31/1899 3:15:00 AM +02:00	On Time	unknown	No		
13	1/1/2024	12/31/1899 4:15:00 AM +02:00	12/31/1899 4:45:00 AM +02:00	12/31/1899 4:45:00 AM +02:00	On Time	unknown	No		
14	1/1/2024	12/31/1899 5:00:00 AM +02:00	12/31/1899 5:30:00 AM +02:00	12/31/1899 5:30:00 AM +02:00	On Time	unknown	No		
15	1/1/2024	12/31/1899 6:30:00 AM +02:00	12/31/1899 7:50:00 AM +02:00	12/31/1899 7:50:00 AM +02:00	On Time	unknown	No		
16	1/1/2024	12/31/1899 7:45:00 AM +02:00	12/31/1899 8:45:00 AM +02:00	12/31/1899 8:45:00 AM +02:00	On Time	unknown	No		
17	1/1/2024	12/31/1899 7:45:00 AM +02:00	12/31/1899 8:45:00 AM +02:00	12/31/1899 8:45:00 AM +02:00	On Time	unknown	No		
18	1/1/2024	12/31/1899 7:30:00 AM +02:00	12/31/1899 8:50:00 AM +02:00	12/31/1899 8:50:00 AM +02:00	On Time	unknown	No		
19	1/1/2024	12/31/1899 7:30:00 AM +02:00	12/31/1899 9:20:00 AM +02:00	12/31/1899 9:20:00 AM +02:00	On Time	unknown	No		
20	1/1/2024	12/31/1899 9:00:00 AM +02:00	12/31/1899 10:50:00 AM +02:00	12/31/1899 10:50:00 AM +02:00	On Time	unknown	No		
21	1/1/2024	12/31/1899 11:15:00 AM +02:00	12/31/1899 12:35:00 PM +02:00	12/31/1899 1:06:00 PM +02:00	Delayed	Technical Issue	Yes		
22	1/1/2024	12/31/1899 10:30:00 AM +02:00	12/31/1899 11:00:00 AM +02:00	12/31/1899 11:00:00 AM +02:00	On Time	unknown	No		
23	1/1/2024	12/31/1899 9:30:00 AM +02:00	12/31/1899 11:20:00 AM +02:00	12/31/1899 11:20:00 AM +02:00	On Time	unknown	No		
24	1/1/2024	12/31/1899 12:45:00 PM +02:00	12/31/1899 2:05:00 PM +02:00	12/31/1899 2:05:00 PM +02:00	On Time	unknown	No		
25	1/1/2024	12/31/1899 1:15:00 PM +02:00	12/31/1899 2:35:00 PM +02:00	12/31/1899 2:35:00 PM +02:00	On Time	unknown	No		

5. Conclusion

This project demonstrates how **Microsoft Power BI** can be used to analyze transportation data and generate valuable insights.

By analyzing UK railway data, the dashboard provides a clear understanding of delays, route efficiency, and travel patterns. These insights can support better decision-making, improve service quality, and enhance passenger experience.